DP-OPS-0048: Restore 'Super' TASK.md as the System Heartbeat, Enforce Worker Exploitation, and Automate Routing Closeout

1. Architectural Philosophy: The Repository as a Metabolic Organism

The trajectory of the Stela repository's architecture has moved decisively toward a model of biological mimicry, specifically in the domains of metabolism and homeostasis. As the system evolves from a static collection of files into a dynamic, self-regulating entity, the mechanisms that govern its state transitions—its "heartbeat"—must be rigorously defined and enforced. The prevailing challenge in this evolutionary phase is the accumulation of "sedimentary context," a form of digital entropy where the artifacts of previous operations contaminate the "Forever Now" of the current execution cycle. This report presents a comprehensive analysis and execution strategy for **DP-OPS-0048**, a pivotal dispatch packet designed to restore the "Super" TASK.md schema, enforce the concept of "Worker Exploitation" through generative verification, and automate the "Routing Closeout" mechanism.

The central thesis of this architectural overhaul is that a repository, much like a biological organism, requires a systole and diastole—a rhythmic contraction of work and a relaxation of cleanup. Without this "heartbeat," the system suffers from the accumulation of "dark data," orphaned artifacts, and stale logic pointers that degrade the fidelity of the Guest Contractor's (the AI agent's) output. The "Close DP Cycle" is not merely an administrative ending to a task; it is the vital pulse that revitalizes the system, purging the toxicity of the past to ensure the purity of the future.

1.1 The "Forever Now" and the Philosophy of Statelessness

The Stela platform operates on a rigid "Filing Doctrine" that partitions the repository into functional zones: ops/ for execution, docs/ for explanation, projects/ for payloads, and storage/ for ephemeral artifacts.[1] However, the efficacy of this doctrine relies on the integrity of the "Forever Now"—the concept that at any given moment, the repository state is the absolute and only truth.

In a truly stateless system, the "current moment" is defined solely by the HEAD commit and the active context files loaded into the agent's working memory. Any information lingering in TASK.md from a previous session—whether it be a completed checklist, a stale git hash, or a drafted commit message—constitutes "Ghost Data." This data appears valid on the surface

but belongs to a defunct timeline. When an agent consumes Ghost Data, it hallucinates a continuity that does not exist, leading to "context drift" where the agent attempts to satisfy the constraints of a past task rather than the demands of the current Dispatch Packet (DP).

To maintain the sanctity of the "Forever Now," the system must practice selective amnesia. The "Super" TASK.md restoration proposed in DP-OPS-0048 is fundamentally an implementation of this amnesia. By automating the scrubbing of TASK.md at the precise moment of closeout, the system enforces a *tabula rasa* state. This ensures that the next worker cannot rely on the "memory" of the file but must re-hydrate the context from the immutable truth of the repository.[1] The cycle of hydration (Open), execution (Work), and dehydration (Closeout/Scrub) constitutes the system heartbeat, a rhythm that prevents the onset of architectural rigor mortis.

1.2 The "Super" TASK.md: A Holographic Command Surface

The "Super" TASK.md is not a passive log file; it is the repository's active prefrontal cortex. As defined in the "Source of Truth Read-in Order," it sits at the critical nexus between the Constitution (PoT.md) and the History (SoP.md).[1] For the system to function with high fidelity, TASK.md must be "Holographic"—meaning it must contain, by reference or inclusion, the entire logic required to execute a task without external dependency or implicit assumptions.

The degradation of the TASK.md structure in recent cycles—specifically the loss of the "Self-Enforcing" sections originally conceptualized in DP-OPS-0045—represents a failure of the system's immune response to entropy. The restoration of the "Super" schema is not an exercise in nostalgia but a reinstatement of necessary complexity. This schema includes distinct functional blocks that map to the cognitive requirements of an autonomous agent:

- **The Anchor (Session State):** Binds the worker to a specific point in spacetime.
- **The Law (Logic Pointers):** Citations of the Constitution that govern behavior.
- **The Contract (Dispatch Packet):** The specific instruction set for the current cycle.
- **The Stream (Work Log):** A chronological record of thought and action.
- **The Seal (Mandatory Closing Block):** The generative verification protocol.

DP-OPS-0048 aims to codify these sections not just as templates but as invariants enforced by the system's heartbeat. If TASK.md does not conform to the "Super" schema at the beginning of a cycle, the system is in a broken state.[1]

1.3 State Hygiene and the Prevention of Dark Data

"Dark Data" refers to artifacts that exist outside the governance of PoT.md—untracked files in /tmp, uncommitted changes in main, or drafting artifacts left in TASK.md.[1] In the current operational model, the reliance on the operating system's standard /tmp directory creates a vector for data leakage and context contamination. These "orphaned" artifacts accumulate unnoticed, occupying storage and potentially confusing future processes that might

inadvertently scan them.

The implementation of System Metabolism Fixes addresses this by redirecting all ephemeral storage to a repository-controlled location: storage/_tmp. This directory is strictly governed by the Filing Doctrine as "Trash" and is excluded via .gitignore to prevent accidental commits.[1] However, unlike the system /tmp, storage/_tmp is within the "reach" of the repository's cleaning tools. The "Routing Closeout" automation acts as the system's lymphatic system, identifying these waste products and neutralizing them before they can metastasize into technical debt. By enforcing strict state hygiene, TASK.md is transformed from a passive dumping ground into a "Living Surface" that actively resists the accumulation of dark data.[1]

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2. The Mechanics of Worker Exploitation: A Cybernetic Paradigm

The term "exploitation" is utilized within this architecture in its strictest cybernetic sense: the extraction of maximum utility from a resource. In the context of Large Language Models (LLMs) acting as Guest Contractors, the "resource" is their token-generation capability and their high-dimensional reasoning capacity. A system that allows a worker to fill out a pre-populated form is under-utilizing that worker; it is wasting the "free labor" of the agent's computation. To ensure high-fidelity operations, the system must "exploit" the worker by forcing it to engage in "Generative Verification."

2.1 Generative Verification vs. Passive Checking

In standard software engineering workflows, a checklist often acts as a passive guardrail. The worker sees [ ] Run Lint and checks it [x] Run Lint. This action requires minimal cognitive load and offers zero proof of execution. The worker can check the box without performing the action, a phenomenon known as "compliance hallucination."

Under the "Enforced Worker Exploitation" model proposed in DP-OPS-0048, the TASK.md closing block is presented as a blank void. The worker is tasked with *constructing* the receipt from scratch. It must query the system state, retrieve the hash, verify the linter output, and then generate the proof string.[1]

- **Passive Interaction:** "Check box if lint passed." (Low cognitive load, Low fidelity)
- **Generative Interaction (Exploited):** "Generate a receipt containing the SHA-256 of the linter output and the timestamp of execution." (High cognitive load, High fidelity)

This requirement forces the model to engage with the underlying reality of the repository. It cannot hallucinate compliance because the "form" of compliance does not exist until the worker creates it. The worker must expend tokens to prove its work, thereby converting its potential energy (reasoning capability) into kinetic energy (verified truth).

2.2 The "Mandatory Closing Block" as a Cognitive Trap

The "Mandatory Closing Block" is explicitly designed as a "Cognitive Trap." It is a barrier that prevents the worker from exiting the session without performing the necessary labor. By creating a template that is *structurally complete* but *semantically empty*, the system creates a vacuum that the worker is compelled to fill.

The user context specifies that this block must never be pre-filled.[1] If ops/bin/open were to populate the closing block with "Expected Result: Pass," the worker would simply adopt that expectation as truth, bypassing the verification step. By scrubbing this section during the Routing Closeout, the system ensures that every "Pass" recorded in history is the result of an active judgment made in the "Forever Now".[1]

Furthermore, the inclusion of the "Varied Wording Provision"—which mandates that the summary in the closing block must differ from the opening statement—defeats the agent's tendency toward "Context Echo." Agents are statistically likely to repeat the input prompt as the output summary. By forbidding this, the system forces the agent to re-process the information, proving that it understands the "Why" of the Dispatch Packet, not just the "What."

2.3 Maximizing Token Utility and Cognitive Density

Every token generated by the AI costs computational resources and introduces a non-zero probability of error. "Exploitation" implies ensuring that every token contributes to the system's fidelity.

- **Low Utility Tokens:** Repetitive boilerplate, polite conversation, restating the prompt, apologies for misunderstandings.
- **High Utility Tokens:** Diff statistics, verification hashes, concise reasoning, error logs, unique summaries.

DP-OPS-0048 optimizes for high-utility tokens by automating the generation of low-utility boilerplate (via ops/bin/open and the TASK.md template) and forcing the worker to focus its output entirely on the unique delta of the current DP. The "Super" TASK.md structure facilitates this by providing rigid slots for high-utility data (e.g., ### 3.4.5 Receipt), ensuring the worker does not waste tokens on formatting or structural invention.[1] This concentration of cognitive density makes the TASK.md a potent artifact of truth.

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3. Structural Analysis: Restoring the 'Super' Schema

The "Super" TASK.md schema, originally defined in **DP-OPS-0045**, serves as the blueprint for this metabolic restoration. A forensic analysis of the current repository state, specifically the dump-full-main-ca9b35f9.txt [1], reveals significant degradation in specific areas, particularly in the "Self-Enforcing" sections. The entropy of manual edits has eroded the rigid structure

required for automated governance.

3.1 Section 1: Session State (The Anchor)

This section anchors the worker in the specific spacetime of the repository. It is the " You Are Here" marker for the agent.

- **Current Status:** Present in the dump but often contains stale data or formatting inconsistencies.
- **Super Requirement:** It must include four specific fields: Pointer, Active Branch, Base HEAD, and Context Manifest.
- **Metabolic Function:** These fields are the variables of the session equation. While ops/bin/open is responsible for populating them at the start of a session, the "Scrub" logic must proactively *unset* them to (none) during closeout. This prevents the "Anchor Mismatch" observed in DP-OPS-0047, where TASK.md referenced an old commit hash (46a04806) while the actual repo head had moved (d14969db).[1] A blank anchor forces the next session to re-establish coordinates.

3.2 Section 2: Logic Pointers (The Law)

This section links the task to the Constitution (PoT.md) and the Toolchain. It acts as the "Miranda Warning" for the AI agent, reminding it of the laws it operates under before it begins work.

- **Status:** Present but often abbreviated.
- **Super Requirement:** It must explicitly list the Jurisdiction (PoT Section 3) and Behavioral Standard (PoT Section 4.2). It must also reference the tools/lint/truth.sh and tools/lint/style.sh scripts as the "Execution Pointers."
- **Metabolic Function:** This section is static in content but dynamic in enforcement. Its presence is required to pass the tools/lint/truth.sh check. The restoration must ensure that the "Read-in Order" (1. PoT, 2. SoP, 3. TASK, 4. MAP) is explicitly cited here to prevent governance drift.

3.3 Section 3: Current Dispatch Packet (The Contract)

This is the core of the work surface, containing the specific instructions for the cycle. The "Super" schema requires a rigid breakdown into sub-sections that mirror the scientific method:

- **3.1 Freshness Gate:** This is a logic puzzle. It lists the Base Branch, Required Work Branch, and Base HEAD. Crucially, it must list the Gate Artifacts (OPEN, PORCELAIN, Dump, Manifest). The worker must verify that the files in storage/handoff/ match these values exactly.
- **3.2 Required Context Load:** An ordered list of files the worker *must* read before generating code. This prevents "blind coding."
- **3.3 Scope and Safety:** Explicit bounds (Allowlist) to prevent drift. This defines the "Hot

Zone" and the files that are off-limits.
- **3.4 Execution Plan:** The A-E Canon (State, Request, Changelog, Patch, Receipt).

Drift analysis shows that TASK.md often loses the explicit "Freshness Gate" detailed validation steps during manual edits. DP-OPS-0048 must hard-code this structure into the TASK.md template used by the ops/bin/open binary.[1]

3.4 Section 4: Closeout (The Verification)

This is the target of the "Worker Exploitation" logic.

- **Current State:** Often reduced to a generic pointer to docs/MANUAL.md.
- **Super State:** A generic, empty block in TASK.md that points to docs/MANUAL.md for *instructions* but requires the *output* to be generated here.
- **The Trap:** The section must contain the "Varied Wording provision" [1], forcing the worker to generate a unique "Final Squash Stub" that differs from the primary commit header. This prevents lazy copying and ensures the worker has synthesized the session's value.

3.5 Section 5: Work Log (The Stream)

The chronological record of the session.

- **Super Requirement:** Timestamped continuity.
- **Metabolism:** This section generates the most "dark data" (intermediate thoughts, failed attempts, debugging logs). The Routing Closeout must harvest valuable insights from here into SoP.md and then mercilessly truncate the rest.[1] The repository cannot bear the weight of every thought ever thought; it only keeps the conclusions.

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4. The Heartbeat: Automating Routing Closeout

The "Heartbeat" is the cycle of data transformation that sustains the repository. It consists of two phases: **Systole** (Contraction/Execution) and **Diastole** (Relaxation/Cleanup). While the execution phase is handled by the worker, the cleanup phase has historically been manual and prone to error. DP-OPS-0048 focuses on automating the Diastole phase via "Routing Closeout."

4.1 The Routing Logic

When a DP is closed, artifacts must be routed to their final resting places. This prevents the repository root and storage/ from becoming a dump. The routing logic is a deterministic protocol:

Table 2: Artifact Routing Protocol

Artifact Type	Origin	Destination	Action	Mechanism
SoP Entry	TASK.md Results	SoP.md	Append	Manual/Agent
Code Changes	work/ branch	main	Merge	Git
Open Artifact	storage/handoff/	storage/archives/context/	Archive	ops/bin/context
Dump Payload	storage/dumps/	storage/archives/context/	Archive	ops/bin/context
Ephemeral TMP	storage/_tmp/	(Void)	Delete	ops/bin/prune
Drafting Text	TASK.md	(Void)	Scrub	ops/bin/prune

4.2 The Semantic Scrub (The Reset)

The most critical innovation in DP-OPS-0048 is the "Semantic Scrub" of TASK.md. This is not a simple file deletion (which would violate the requirement for TASK.md to exist) nor a complete overwrite (which might destroy structural pointers and git history). It is a surgical removal of content while preserving the schema.[1]

The implementation relies on **GNU awk (gawk)**, which allows TASK.md to be treated as a text-based state machine. The scrub script will parse the file line by line, identifying specific block headers (## 3. Current Dispatch Packet, ## 5. Work Log) and applying transformational logic:

1. **Identify Blocks:** Use regex to find the start of the DP section and the Work Log.
2. **Neutralize:** Replace specific lines (e.g., Active Branch:...) with (none) or blank templates.
3. **Truncate:** Delete all text following the ## 5. Work Log header, leaving only the header itself and a placeholder like (No active thread).
4. **Sanitize:** Revert the DP Header to a generic template ### DP-XXXX: (Template).

This ensures that when the next worker runs ops/bin/open, they are presented with a clean structure that *demands* to be filled. The worker cannot inherit the previous worker's context; they must establish their own.[1]

4.3 Redirection of Ephemeral Storage

To support the "Heartbeat," the system must also control its waste products. Currently, ephemeral artifacts may be scattered or rely on the system-wide /tmp directory. Utilizing /tmp creates "orphaned" artifacts that exist outside version control, leading to "dark data" accumulation and potential security risks like race conditions (TOCTOU).[1]

DP-OPS-0048 enforces the redirection of all temporary outputs to storage/_tmp.

- **Implementation Strategy:** Use the mktemp command with the TMPDIR environment variable set inline: tmp_file=$(TMPDIR="storage/_tmp" mktemp -t stela-prune.XXXXXX).
- **Hygiene:** storage/_tmp is explicitly defined in PoT.md as "Trash" and is excluded via .gitignore.
- **Lifecycle:** The ops/bin/prune script will be engineered to purge this directory upon successful completion of the Closeout cycle. If the script fails, the artifacts remain for forensic inspection, providing "localized, controlled persistence".[1]

4.4 Automating Revitalization

The heartbeat is completed when the system is ready for the next pulse. This requires the "Routing Closeout" to not only clean up but to *prep* the environment.

- **LLMS Refresh:** The ops/bin/llms binary must be triggered to regenerate the context bundles (llms.txt, llms-full.txt). This ensures that the new "Forever Now" state (post-merge) is immediately reflected in the AI's discovery maps.[1]
- **Map Update:** ops/bin/map must run to update the auto-generated index in docs/MAP.md.

By automating these steps within the transition script (triggered by ops/bin/prune or a wrapper), the system minimizes the "Time to Freshness" for the next operator.

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5. Protection of the Truth: Governance and Legislation

Automation without legislation is merely a script; automation backed by PoT.md is governance. To "Protect the Truth" as requested, the definition of TASK.md, the Closeout Cycle, and the Metabolic Law must be enshrined in the Policy of Truth.[1]

5.1 Amendment: The Metabolic Law

We will introduce a new Axiom to **Section 1.2** of PoT.md to formalize the heartbeat:

> **Metabolic Law:** The repository operates in a "Forever Now." TASK.md must be scrubbed to a neutral, template state at the conclusion of every Dispatch Packet cycle. Reliance on stale TASK.md data from previous sessions is a system failure.

The "Routing Closeout" is mandatory.

5.2 Amendment: The Generation Mandate

We will introduce a directive to **Section 4.2 (Behavioral Logic Standard)** to enforce worker exploitation:

> **Generation Mandate:** Worker verification must be generative. Pre-filled completion checks in TASK.md are forbidden. Workers must populate blank templates with fresh proofs extracted from the immediate session context. Any template that provides a default "Pass" or "Checked" state is a violation of this standard.

5.3 Codifying the "Super" Structure

The structure of TASK.md will be protected by referencing it in PoT.md Section 4.3 (Hard Constraints). The tools/lint/dp.sh script will be updated to validate that the file *always* contains the five required sections (Session State, Logic Pointers, Current DP, Closeout, Work Log). If a worker attempts to submit a PR that deletes one of these sections (e.g., removing the Work Log to save tokens), the linter will reject it, enforcing the "Holographic" nature of the file.

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6. Execution Plan: DP-OPS-0048 Implementation

This section defines the operational steps to execute DP-OPS-0048. This plan adheres to the A-E Canon of the "Super" schema.

6.1 State (A) - The Baseline

- **Current Status:** TASK.md is active but degraded (Anchor Mismatch, missing sections). ops/bin/prune lacks scrubbing logic. PoT.md lacks metabolic definitions. storage/_tmp does not exist or is not enforced.
- **Risk:** High probability of context hallucination in the next session due to stale data in TASK.md. "Dark data" accumulation in system /tmp.

6.2 Request (B) - The Objective

1. **Refactor TASK.md:** Restore the full "Super" schema from DP-OPS-0045, ensuring all 5 sections are present and "Self-Enforcing."
2. **Hardening:** Convert all "Closeout" fields to blank templates to enforce Worker Exploitation.
3. **Automation:** Upgrade ops/bin/prune to include the scrub_task function (gawk logic) and storage/_tmp cleanup.
4. **Legislation:** Update PoT.md with Metabolic Law and Generation Mandate.

5. **Infrastructure:** Create storage/_tmp and update .gitignore.

6.3 Execution (C) - The Implementation Strategy

The operator will perform the following mutations:

1. Infrastructure Setup:

Bash

```
mkdir -p storage/_tmp
echo "*" > storage/_tmp/.gitignore
echo "!.gitignore" >> storage/_tmp/.gitignore
```

2. ops/bin/prune Upgrade:

Modify the script to accept a --scrub flag (or run by default after archiving). Implement the scrub_task function using gawk to semantically reset TASK.md.

Bash

```
# Conceptual Logic for ops/bin/prune extension
scrub_task() {
  echo "Scrubbing TASK.md for heartbeat reset..."
  gawk -i inplace '
    BEGIN { printing=1 }
    /^## 1. Session State/ { print; print "Active Branch: (none)"; print "Base HEAD: (none)"; print "Context Manifest: ops/lib/manifests/CONTEXT.md"; printing=0; next }
    /^## 2. Logic Pointers/ { printing=1 }
    /^### 3.1 Freshness Gate/ { print; print "Base Branch: (none)"; print "Base HEAD: (none)"; printing=0; next }
    /^## 3.2 Required Context Load/ { printing=1 }
    /^### DP-/ { print "### DP-XXXX: (Template)"; next }
    /^## 5. Work Log/ { print; print "(No active thread)"; exit }
    printing { print }
  ' "$TASK_FILE"
}
```

3. TASK.md Restoration:

Manually reset the file to the "Super" state. This involves clearing all current DP data, setting fields to (none), and ensuring the "Mandatory Closing Block" contains the specific "Varied Wording" instructions but *no* pre-filled content.

4. PoT.md Update:

Insert the text defined in Section 5.1 and 5.2.

6.4 Verification (D)

- **Test 1 (Heartbeat):** Execute ./ops/bin/prune. Verify TASK.md is scrubbed to the template state. Verify storage/_tmp is empty.
- **Test 2 (Exploitation):** Attempt to commit a TASK.md with pre-filled checkboxes. Verify tools/lint/dp.sh (if updated) or manual review rejects it.
- **Test 3 (Routing):** Confirm that ops/bin/open correctly populates the *fresh* TASK.md after a scrub.

6.5 Receipt (E)

The final artifact of DP-OPS-0048 will be a SoP.md entry confirming that the "Heartbeat" mechanism is active and TASK.md has been successfully revitalized.

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7. The "Forever Now": Implications and Ripple Effects

The successful implementation of DP-OPS-0048 establishes the "Forever Now" as the operative reality of the Stela repository.

7.1 Elimination of Hallucination Vectors

LLMs often hallucinate based on "few-shot" examples present in their context window. If TASK.md contains the completed log of the *previous* task, the model is statistically likely to mimic the patterns, file paths, and decisions of that previous task, even if they are irrelevant to the current one. By scrubbing TASK.md, we remove these "distractor shots." The model is forced to rely on the immutable docs/ and the current SoP.md for its patterns, ensuring that its behavior is derived from the *constitution*, not the *recent past*.[1]

7.2 Second-Order Effect: Audit Trail Purity

Because the worker is forced to generate the "Final Squash Stub" and "Review Conversation Starter" from scratch (Worker Exploitation), these artifacts become high-fidelity signals of the worker's internal state. If the worker fails to understand the work it just performed, it will fail to generate a coherent Squash Stub. This acts as a secondary "Turing Test" for the work unit

itself. If the meta-data is garbage, the code payload is likely suspect.

7.3 Third-Order Effect: System Resilience

The "Heartbeat" creates a system that is resilient to crashes. Because the "Resting State" of the system is explicitly defined (Blank TASK.md, Clean storage/_tmp), any interrupted session can be recovered by simply running ops/bin/open or ops/bin/prune. There is no "partial state" to unwind because the system enforces a hard reset between cycles. This transforms the repository from a fragile stateful entity into a robust, restartable service.[1]

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8. Conclusion: The Revitalization of Stela

DP-OPS-0048 is not merely a maintenance task; it is a constitutional amendment to the physics of the Stela repository. By restoring the "Super" TASK.md schema, the system regains its prefrontal cortex—a centralized, comprehensive command surface capable of directing complex operations. By enforcing "Worker Exploitation," the system ensures that this surface is populated with high-utility, verified truth rather than passive hallucinations. Finally, by automating the "Routing Closeout" via the ops/bin/prune heartbeat, the system achieves the "Forever Now"—a state of perpetual readiness, unburdened by the ghosts of previous cycles.

The execution of this plan will transition Stela from a passive repository into a metabolic organism, capable of self-cleaning and self-verification, thereby securing its operational integrity against the entropy of autonomous development. The system is no longer just a place where code lives; it is a system that lives.

Works cited

1. dump-full-main-ca9b35f9.txt